Local-View-Assisted Discriminative Band Selection With Hypergraph Autolearning for Hyperspectral Image Classification

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***Abstract*— For hyperspectral images (HSIs), it is a challenging task to select discriminative bands due to the lack of labeled samples and complex noise. In this article, we present a novel local-view-assisted discriminative band selection method with hypergraph autolearning (LvaHAl) to solve these problems from both local and global perspectives. Specifically, the whole band space is first randomly divided into several subspaces (LVs) of different dimensions, where each LV denotes a set of low-dimensional representations of training samples consisting of bands associated with it. Then, for different LVs, a robust hinge loss function for isolated pixels regularized by the row-sparsity is adopted to measure the importance of the corresponding bands. In order to simultaneously reduce the bias of LVs and encode the complementary information between them, samples from all LVs are further projected into the label space. Subsequently, a hypergraph model that automatically learns the hyperedge weights is presented. In this way, the local manifold structure of these projections can be preserved, ensuring that samples of the same class have a small distance. Finally, a consensus matrix is used to integrate the importance of bands corresponding to different LVs, resulting in the optimal selection of expected bands from a global perspective. The classification experiments on three HSI data sets show that our method is competitive with other comparison methods.**

***Index Terms*— Auto-learning hypergraph, capped hinge loss (HL) function, hyperspectral image (HSI) classification, local-view-assisted learning, row-sparsity constraint, supervised band selection (BS).**

I. Introduction

With the development of imaging techniques, hyperspectral sensors are capable of deepening the characterization of various objects with hundreds of contiguous bands. Each hyperspectral pixel is a set of reflection values that are related to the physical, chemical, and optical properties of the material itself, which lays the foundation of labeling pixels, namely, hyperspectral images (HSIs) classification [1], [2], and benefits many practical applications [3]–[5]. For classification, a wealth of spectral bands not only increases the computational and storage burden of training a classifier but may also degrade the classification accuracy. For instance, due to the lack of labeled pixels, the generalization capability of the classifier is limited when high-dimensional bands are fed back, namely, the curse of dimensionality [6]. In addition, many adjacent bands may be heavily redundant and fail to provide additional discriminative information [7].

Reducing the number of bands, that is, dimensionality reduction [8]–[12], is an effective strategy to solve the aforementioned challenges. In the field of HSI, it can be implemented by three techniques, including feature extraction (FE) [13]–[15], unmixing [16]–[18], and band selection (BS). BS has the following three advantages over the other two techniques. First, it only obtains a subset of the original bands and does not generate new features, thereby preserving the physical information from the selected bands [11]. Second, the FE and unmixing techniques typically need all test samples to extract new features or endmembers and corresponding abundances during the test phase. Compared with them, BS only stores the information related to a few selected bands, which greatly reduces the storage and computational burden. Third, BS can be combined with the FE and unmixing techniques to improve the efficiency and performance of the latter, as done in [19]–[21].

According to whether the label information is used, BS is usually performed in two ways: unsupervised and supervised. Due to the absence of labels, representative bands are mined by measuring the properties of data itself in the unsupervised methods. A variety of metrics have been designed over the past few years, including variance-based method [22], clustering-based method [23], gradient-based method [24], ranking-based method [25], evolutionary-based method [26], and geometry-based method [27]. However, the bands selected by these methods may lack discriminative information that simultaneously distinguishes multiple classes, resulting in poor classification performance. Benefiting from the prior knowledge of labels, the discriminative bands can be selected by wrapper- and filter-based supervised methods. In the wrapper-based methods, if a band is included in the current candidate subset and the resulting subset enables the training samples to have a maximum cross-validation accuracy associated with a given classifier, then this band will be selected. The above-mentioned process is repeated until the required number

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of bands is met [28], [29]. The filter-based methods focus on designing criteria that examine the relationship between various band combinations and labels to select discriminative bands. For this purpose, several indexes have been proposed from different perspectives, such as distance-based index [30], mutual information-based index [31], and class separability-based index [32]. Fisher score (FS) [33], as a classic class separability measure, is dedicated to discovering a band subset, such that in the low-dimensional space spanned by the selected bands, samples in the same class are as close as possible, while samples in different classes are as far away as possible. These methods measure different band subsets in the original band space, which may not effectively capture a nonlinear relationship between classes. To overcome this challenge, Damodaran *et al*. [34] proposed a new index to examine discriminant bands in the reproducing kernel Hilbert space.

With the popularity of matrix computing, BS is performed by solving a variety of matrix-based optimization objectives consisting of different loss functions and regularization terms (e.g., low rank and sparsity). For example, Wang *et al*. [35] used the low-rank constraint on the block-diagonal matrix to describe the correlation characteristics among bands. Zhu *et al*. [36] captured the interband redundancy by means of low-rank modeling. However, these low-rank-based models typically require an additional clustering to obtain distinct bands, which falls into a dilemma of balancing the informativeness and distinctiveness of selected bands [37]. The sparse norm of a vector or matrix can perform a kind of continuous variable selection, avoiding the extra selection process [38], [39]. Hence, sparse constraints have been widely used for BS. Sun *et al*. [40] formulated BS into a linear construction with l_1-norm constraint on the coefficient matrix and proposed a BS method based on the sparse subspace. To reduce the dependence between selected bands, a dissimilarity-weighted regularization incorporating multiple dissimilarity measurements was further presented to encode the differences between pairwise bands [41]. Zhang *et al*. [42] combined the ϵ-draggings and least square regression model regularized by $l_{2,1}$-norm to enlarge the distance between pixels from different classes and select discriminative bands. In order to obtain bands that are positively correlated with the classification results, Yang and Bao [43] combined the log-likelihood function and l_1-norm to implement BS and classification simultaneously. Wei *et al*. [44] utilized the hinge loss (HL) constrained by $l_{2,1}$-norm to select bands that maximize the margin between classes.

In HSIs, the reflectance spectra of each pixel are affected by several factors, such as illumination, moisture, and other adjacent pixels in space, which is known as "endmember/spectral uncertainty" [45]. Hence, we often encounter isolated pixels, e.g., they have spectra that are significantly different from other pixels with the same label. The unmixing is an effective solution to this situation, which mainly includes iterative mixture analysis, spectral BS, spectral weighting bands, spectral transformations, and spectral model [46]. Soltanifarani *et al*. [16] formulated the unmixing into a dictionary learning model and then imposed a row-sparsity constraint on the abundance matrix to mitigate the effects of endmember uncertainty. For the same propose, Wang *et al*. [47] proposed an unmixing method based on the nonnegative matrix factorization with spatial group sparsity by combining spatial information. However, setting the appropriate number of endmembers is a challenge in these methods. Hence, different attempts to mitigate noise have been studied when performing BS. For example, Sun and Du [48] utilized a robust principal component analysis to estimate the clean low-rank approximation of a band matrix and remove the noise components. The Laplacian graph was incorporated to further improve the performance of BS. From the perspective of considering the quality of samples, Wei *et al*. [49] used a weight learning mechanism that is positively correlated with sample quality to deal with different samples and improve the robustness to noise. However, none of these methods take into account the conflict between small labeled samples and high-dimensional bands. Besides, they lack a mechanism to specifically deal with isolated pixels when their labels are incorrectly estimated in BS.

Hence, we propose a supervised local-view-assisted BS method with hypergraph autolearning (LvaHAl) when only a few labeled samples are available in this article. In the LvaHAl method, the whole band space is first divided into several subspaces with different dimensions. Each band subspace corresponds to an LV. Hence, we can obtain multiple representations of samples consisting of different band subsets. Although any method of generating LVs (e.g., clustering) is feasible, random dividing is adopted in our article because of its simplicity and speed. In this way, we can obtain a large number of low-dimensional representations of training samples, which may alleviate the instability of selected bands caused by the lack of labeled samples in supervised methods [50]. Then, for each LV, considering the isolated pixels, a novel measure function consisting of HL with an upper bound and a row-sparsity term is designed to evaluate the bands associated with this view. In order to simultaneously reduce the bias of LVs and encode the complementary information between them, samples from all LVs are further projected into the label space. Subsequently, a new hypergraph model is designed to locally align projections from the same class. It not only allows different LVs to share the same structural distribution but also preserves the specificity of each LV by automatically adjusting the view-dependent hyperedge weights. Finally, based on the assumption that the importance of bands is continuous [49], a consensus matrix that records the importance of all bands is regularized to integrate the importance of bands associated with different LVs and select globally optimal bands. The contributions of this article are summarized as follows.

1) We propose a novel BS method LvaHAl to solve the conflict between insufficient labeled samples and high-dimensional bands. This method can obtain a large number of low-dimensional representations of labeled samples by dividing the band space. An alternate iterative algorithm is developed to solve the LvaHAl model. Besides, an online version of LvaHAl is developed to enable it to process dynamic data.
2) A nonconvex and nonsmooth measure function that is theoretically robust to isolated pixels is presented to estimate the importance of bands. It is formulated as

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HL with an upper bound regularized the row sparsity, which can be solved by a well-designed algorithm.

3) A novel information-sharing mechanism based on a hypergraph is presented. It allows various LVs to share the same structural distribution while preserving the specificity of each LV by automatically adjusting the view-dependent hyperedge weights, which is more general and flexible than the traditional graph-based manner.

The remainder of this article is organized as follows. Section II describes the background knowledge of basic notations and hypergraph learning. Section III presents the details of our proposed BS method. The experimental results are presented and analyzed in Section IV. The conclusion is summarized in Section V.

II. Background Knowledge

A. Basic Notations

Before processing, the basic notations used throughout this article are stated. Vectors and matrices are represented by boldface lowercase letters and boldface uppercase letters, respectively. For a given matrix $\mathbf{M} = (m_{\mathrm{ij}}) \in \mathbb{R}^{m\times n}$, the ith row and jth column are denoted by $\mathbf{m}^i$ and $\mathbf{m}_j$, respectively. The l_0-norm of vector $\mathbf{v} = (v_i) \in \mathbb{R}^m$ is represented by $\|\mathbf{v}\|_0$, which is defined as the number of nonzero elements. The corresponding l_1-norm and l_2-norm are defined as $\|\mathbf{v}\|_1 = \sum_i |v_i|$ and $\|\mathbf{v}\|_2 = (\sum_i v_i^2)^{1/2}$, respectively. The Frobenius norm of matrix $\mathbf{M}$ is denoted by $\|\mathbf{M}\|_F = (\sum_{i,j} m_{ij}^2)^{1/2}$. The $l_{2,1}$-norm of matrix $\mathbf{M}$ is defined as

$$\|\mathbf{M}\|_{2,1} = \sum_{i=1}^{m} \sqrt{\sum_{j=1}^{n} m_{ij}^2} = \sum_{i=1}^{m} \|\mathbf{m}^i\|_2. \tag{1}$$

If $\mathbf{M}$ is a square matrix, namely, $m = n$, its trace is defined as $\mathrm{tr}(\mathbf{M}) = \sum_i m_{\mathrm{ii}}$. The transpose of matrix $\mathbf{M}$ is denoted by $\mathbf{M}^{\mathrm{T}}$. The symbol $\mathbf{I}$ represents an identity matrix whose dimension is determined according to the context. We let $\mathbf{1}$ denote the all ones vector or matrix. Besides, the expression $\mathbf{M} \geq 0$ or $\mathbf{v} \geq 0$ indicates that all elements of $\mathbf{M}$ or $\mathbf{v}$ are equal or greater than 0.

Given n training pixels $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$, where d is the number of bands, the label vector corresponding to $\mathbf{x}_i$ is represented by $\mathbf{y}_i \in \mathbb{R}^C$. For each $\mathbf{x}_i$, the cth element of $\mathbf{y}_i$ is 1 if $\mathbf{x}_i$ is labeled as class c, and is -1 otherwise. The sample matrix of the vth LV is denoted by $\mathbf{X}_{(v)} = [\mathbf{x}_1^{(v)}, \ldots, \mathbf{x}_n^{(v)}]$, where the ith column $\mathbf{x}_i^{(v)} \in \mathbb{R}^{d_v}$ represents the ith sample with $d_v (d_v < d)$ bands. The major notations used in this article are summarized in Table I.

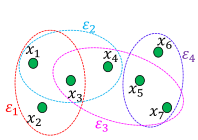
B. Hypergraph Learning

Although undirected graphs have been extensively studied in HSI analysis, including BS [48], FE [51], and unmixing [52], it only describes the relationship between pairwise vertexes. Hypergraph learning [53]–[55], as a generalization of graphs, has the ability to simultaneously characterize the high-order relationship of multiple vertexes.

Let $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ be a hypergraph, where $\mathcal{V} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ is a set of vertexes and $\mathcal{E} = \{\mathcal{E}_1, \ldots, \mathcal{E}_{|\mathcal{E}|}\}$ is a collection

TABLE I
Major Notations

Notations	Descriptions
d	Total number of bands
d_v	The number of bands of the v-th LV
n	The number of training samples
V	The number of LVs
C	The number of classes
r	The weight redistribution parameter
$\epsilon_{(v)}$	The maximum loss of samples in the v-th LV
$\mathbf{x}_i \in \mathbb{R}^d$	The i-th sample with all bands
$\mathbf{q}_{(v)} \in \mathbb{R}^C$	The hyperedge weight vector in the v-th LV
$\mathbf{v}_{(v)} \in \mathbb{R}^n$	The vertex degree vector in the v-th LV
$\mathbf{b}_{(v)} \in \mathbb{R}^n$	The bias vector in the v-th LV
$\mathbf{x}_i^{(d_v)} \in \mathbb{R}^{d_v}$	The i-th sample with d_v bands in the v-th LV
$\mathbf{e}_{(v)} \in \mathbb{R}^C$	The hyperedge degree vector in the v-th LV
$\mathbf{X}_{(v)} \in \mathbb{R}^{d_v\times n}$	Sample matrix in the v-th LV
$\mathbf{W}_{(v)} \in \mathbb{R}^{d_v\times C}$	The projection matrix in the v-th LV
$\mathbf{M}_{(v)} \in \mathbb{R}^{C\times n}$	The nonnegative slack matrix in the v-th LV
$\mathbf{H} \in \mathbf{R}^{n\times C}$	The common incident matrix for all LVs
$\mathbf{W} \in \mathbb{R}^{d\times C}$	The consensus matrix



	ε_1	ε_2	ε_3	ε_4
x_1	1	1	0	0
x_2	1	0	0	0
x_3	1	1	1	0
x_4	0	1	1	0
x_5	0	0	1	1
x_6	0	0	0	1
x_7	0	0	1	1

(a) (b)

Fig. 1. Hypergraph example. (a) Visual structure. (b) Corresponding incident matrix.

of hyperedges. $|\mathcal{E}|$ is the total number of hyperedges. In the hypergraph, each hyperedge $\mathcal{E}_i (1 \leq i \leq |\mathcal{E}|)$ is a nonempty subset of the set $\mathcal{V}$, and the union of all hyperedges is equal to the vertex set. Different methods of constructing hyperedges extract various high-level information. For example, with the help of labels, vertexes with the same label can be included in the same hyperedge. Denote the incident matrix $\mathbf{H}$ as a $n \times |\mathcal{E}|$ matrix, whose entry h_{ij} is 1 if $\mathbf{x}_i \in \mathcal{E}_j$ and is 0 otherwise. The value of $|\mathcal{E}|$ is equal to the number of classes when each hyperedge represents a class. Fig. 1 shows an example of a hypergraph. We let $\mathbf{q} = (q_i) \in \mathbb{R}^{|\mathcal{E}|}$ denote the weight vector of hyperedges, where the ith element q_i corresponds to the weight of $\mathcal{E}_i$. Similarly, let $\mathbf{v} = (v_i) \in \mathbb{R}^n$ and $\mathbf{e} = (e_i) \in \mathbb{R}^{|\mathcal{E}|}$ represent the degree vectors of vertexes and hyperedges, respectively. By these definitions, the degree v_i of the vertex $\mathbf{x}_i$ and the degree e_j of the hyperedge $\mathcal{E}_j$ can be calculated, respectively, by

$$v_i = \sum_{j=1}^{|\mathcal{E}|} q_j h_{ij} \quad \text{and} \quad e_j = \sum_{i=1}^{n} h_{ij}. \tag{2}$$

For given training samples $\{\mathbf{x}_i\}_{i=1}^n$, let $\mathbf{a}_i$ denote a new representation of $\mathbf{x}_i$. The local manifold structure based on the hypergraph claims that the representations corresponding to samples from the same hyperedge should be similar and their similarity is proportional to the hyperedge weight, which can be achieved by optimizing the following model:

$$\min_{\mathbf{A}} \sum_{\mathcal{E}_c\in\mathcal{E}} \sum_{\mathbf{x}_i,\mathbf{x}_j\in\mathcal{E}_c} \|\mathbf{a}_i - \mathbf{a}_j\|_2^2 \frac{q_c}{2\ e_c} = \mathrm{tr}(\mathbf{ALA}^{\mathrm{T}}) \tag{3}$$

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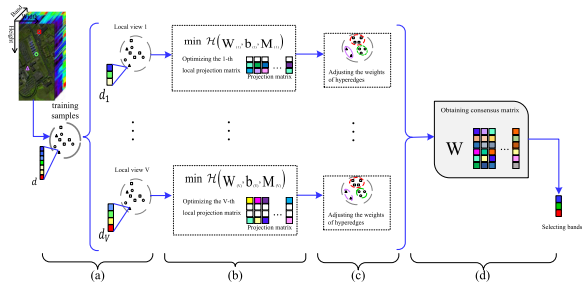


Fig. 2. Flowchart of the proposed LvaHAl method with four components. (a) Generating several LVs for given labeled training samples. Points of the same shape represent samples from the same class. (b) Evaluating some bands associated with different LVs using a robust measure function for the isolated pixels. (c) Sharing information between LVs through a hypergraph that can adjust the view-dependent weights of hyperedges. Circles of different colors (red, green, and pink) represent hyperedges associated with different classes, and the width of each circle reflects the weight of the corresponding hyperedge. (d) Integrating local projection matrices from various LVs and obtaining the consensus matrix to select the expected bands from a global perspective.

where $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n]$. The unnormalized hypergraph Laplacian matrix $\mathbf{L}$ can be formalized as

$$\mathbf{L} = \mathbf{V} - \mathbf{HQE}^{-1}\mathbf{H}^{\mathrm{T}}. \tag{4}$$

In (4), $\mathbf{Q}$, $\mathbf{V}$, and $\mathbf{E}$ represent the diagonal matrices, where the ith diagonal element corresponds to q_i, v_i, and e_i, respectively. In summary, learning a suitable hypergraph is important to preserve the local manifold structure. By analyzing the definition of hypergraph, we can find that the hypergraph learning involves two key problems: 1) the construction of hyperedges and 2) the calculation of the weight of each hyperedge.

III. Proposed Model for Band Selection

Fig. 2 shows the overall framework of the proposed LvaHAl method consisting of four parts: generating LVs that consist of different band subsets, evaluating the bands associated with different LVs, sharing information between LVs, and obtaining the consensus matrix that integrates band information from all LVs to select expected bands. The details of the above-mentioned four parts are described in the following.

A. LvaHAl Method

1) Generating LVs: In supervisory-based model learning, the conflict between a small number of training samples and high-dimensional bands may lead to biased estimation of model parameters, which will result in some meaningless bands being selected when performing BS, further hindering the performance of subsequent processing (e.g., classification). The proposed LvaHAl method solves the conflict by generating LVs. In detail, given a set of training samples $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$ with labels $\{\mathbf{y}_i \in \mathbb{R}^C\}_{i=1}^n$ from C classes, where n and d represent the total number of samples and bands, respectively. When the band dimension is limited between k_1 and k_2, we can theoretically obtain $\sum_{k_1 \le k \le k_2} \binom{d}{k}$ LVs by randomly dividing the whole band space. In other words, we have $n \times \sum_{k_1 \le k \le k_2} \binom{d}{k}$ low-dimensional training samples with some bands related to different LVs. There are two aspects of attention to LVs. First, the intersection of band subsets corresponding to different LVs may not be empty, i.e., a band can appear in multiple LVs. Second, each band is available in at least one LV to ensure that it has a probability of being selected.

Although some sophisticated methods (e.g., clustering) can be used, we use the method of randomly dividing band space as defined in the following to obtain LVs—first, the dimension of the υth LV is determined by the following formula:

$$d_\upsilon = \lfloor [(1-\sigma_1)\tau_{\min} + \sigma_1 \tau_{\max}] \times B \rfloor \tag{5}$$

where d_υ represents the number of available bands for the υth LV, and $\sigma_1 \in [0, 1]$ is a uniform random variable. Second, the band associated with this LV is selected one by one, whose index is determined by the following formula:

$$\mathrm{ind} = \lfloor 1 + \sigma_2 \times B \rfloor \tag{6}$$

where ind represents the index of the selected band, and $\sigma_2 \in [0, 1]$ is a uniform random variable. This step is repeated until d_υ bands are selected for the υth LV. The above-mentioned process of generating LVs is repeated until all bands appear in at least one of LVs. In this way, we can get a large number of labeled low-dimensional samples.

2) Evaluating the Bands of the υth LV: Without loss of generality, the training samples from the υth LV are represented by the matrix $\mathbf{X}_{(\upsilon)} = [\mathbf{x}_1^{(\upsilon)}, \ldots, \mathbf{x}_n^{(\upsilon)}] \in \mathbb{R}^{d_\upsilon \times n}$, where each column denotes a sample with d_υ bands. In the matrix computing paradigm, discriminative BS can be stated as finding a projection variable that involves only a few bands to make the training samples separable in the label space. In other words, if a sample constructed by discriminative bands is correctly classified, the corresponding loss should be zero. To clearly describe the principle of the LvaHAl model, we start with the binary class case. Taking the υth LV as an example, a specific BS using HL, in this case, can be defined as

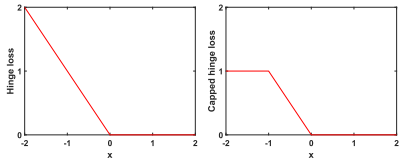
$$\begin{aligned} &\min_{\mathbf{w},b} \sum_{i=1}^{n} [1 - y_i(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i^{(\upsilon)} + b)]_+ \\ &\text{s.t. } \|\mathbf{w}\|_0 = k \end{aligned} \tag{7}$$

where $y_i \in \{+1, -1\}$ is the label of $\mathbf{x}_i^{(\upsilon)}$, k is the number of selected bands, b is the bias term, and $\mathbf{w} \in \mathbb{R}^d$ is the projection vector we want to learn. The HL $\mathcal{L} = (1 - \cdot)_+$ is defined as $\max(1 - \cdot, 0)$. Compared with the l_2-norm-based loss functions [39], [42], the HL of $\mathbf{x}_i^{(\upsilon)}$ can be exactly zero when it is correctly classified. Besides, the l_0-norm imposed on $\mathbf{w}$ guarantees that only k bands play a role in determining the label. The problem (7) is difficult to solve directly. Hence, it can be relaxed to the following sparse (SSVM) problem [38]:

$$\min_{\mathbf{w},b} \sum_{i=1}^{n} \left[1 - y_i\left(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i^{(\upsilon)} + b\right)\right]_+ + \lambda \|\mathbf{w}\|_1 \tag{8}$$

where λ is a regularization parameter that controls the sparseness of $\mathbf{w}$. In unconstrained optimization (8), the l_1 norm of $\mathbf{w}$ plays a similar role to the constraint in objective (7). However, the model (8) has two shortcomings. First, it is designed specifically for the binary class problem. Although both one-versus-one and one-versus-rest enable it to handle multiclass

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(a) (b)

Fig. 3. (a) HL. (b) Capped HL with an upper bound 1.

case, they also cause ambiguity problem while increasing the computational burden. Second, HL is not robust enough to the isolated pixels, as shown in Fig. 3(a). Fig. 3(a) indicates that HL cannot limit the loss caused by the misclassification of isolated pixels. For the second problem, a reasonable strategy is that when the isolated pixels are misjudged, there should be an upper bound on the corresponding loss. Then, a robust capped HL [56] can be defined as follows:

$$\mathcal{L}_c = \min(\mathcal{L}, \epsilon) \tag{9}$$

with an upper bound $\epsilon > 0$, as shown in Fig. 3(b). It is not difficult to find that $\mathcal{L}_c$ is equivalent to $\mathcal{L}$ if $\epsilon \to +\infty$. Xiang *et al.* [57] showed that the HL of sample $\mathbf{x}_i^{(v)}$ in models (7) and (8) can be equivalently expressed as follows:

$$\mathcal{L} = \min_{m_i \geq 0} \left| \mathbf{w}^{\mathrm{T}} \mathbf{x}_i^{(v)} + b - y_i - y_i m_i \right| \tag{10}$$

with a slack variable m_i. Therefore, the capped HL of sample $\mathbf{x}_i^{(v)}$ for multiclass problem can be extended as follows:

$$\mathcal{L}_c = \min \left(\min_{\mathbf{m}_i^{(v)} \geq 0} \left\| \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_i^{(v)} + \mathbf{b}_{(v)} - \mathbf{y}_i - \mathbf{y}_i \odot \mathbf{m}_i^{(v)} \right\|_2^2, \epsilon_{(v)} \right) \tag{11}$$

where $\mathbf{W}_{(v)} \in \mathbb{R}^{d_v \times C}$, $\mathbf{b}_{(v)} \in \mathbb{R}^C$, and $\epsilon_{(v)} \in \mathbb{R}$ are the local projection matrix, the bias vector, and the upper bound of the loss for the vth LV, respectively. The slack vector $\mathbf{m}_i^{(v)} \in \mathbb{R}^C$ is used to encode the loss of sample $\mathbf{x}_i^{(v)}$, and the cth element of $\mathbf{y}_i \in \mathbb{R}^C$ is 1 if $\mathbf{x}_i^{(v)}$ belongs to class c, and is -1 otherwise. Hence, the BS model based on the capped HL for the vth LV can be formulated as follows:

$$\begin{aligned} &\min \mathcal{F}(\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}) \\ &\quad = \sum_{i=1}^{n} \min \left(\left\| \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_i^{(v)} + \mathbf{b}_{(v)} - \mathbf{z}_i^{(v)} \right\|_2^2, \epsilon_{(v)} \right) \\ &\quad + \lambda \|\mathbf{W}_{(v)}\|_{2,1} \\ &\text{s.t. } \mathbf{m}_i^{(v)} \geq 0, \quad i = 1, \ldots, n \end{aligned} \tag{12}$$

with $\mathbf{z}_i^{(v)} = \mathbf{y}_i + \mathbf{y}_i \odot \mathbf{m}_i^{(v)}$, $\mathbf{M}_{(v)} = [\mathbf{m}_1^{(v)}, \ldots, \mathbf{m}_n^{(v)}]$, and λ is a nonnegative regularization parameter. According to (1), $\|\mathbf{W}_{(v)}\|_{2,1}$ can theoretically make the values of some rows of matrix $\mathbf{W}_{(v)}$ zero, which means that the corresponding bands are meaningless in determining the labels of training samples. Conversely, the larger the l_2-norm of the ith row of matrix $\mathbf{W}_{(v)}$, the better the discriminability of the corresponding band. Hence, the model (12) has the ability to evaluate and select the discriminative bands for the vth LV.

3) Hypergraph-Based Information Sharing: Obviously, the model (12) deals with all LVs separately and measures the importance of different bands from one side. The key task of solving the one-sidedness is to achieve information sharing between LVs while maintaining their specificity. Two aspects of knowledge can be mined by analyzing various LVs. First, LVs reflect different representations of given training samples. In other words, representations from different LVs have the same structural distribution. For example, if two representations belong to the same class in one of LVs, they belong to the same class in other LVs. Second, the correlation between representations is view dependent. For example, the representations of the two training samples in one of LVs are highly correlated; however, this may not be true in other LVs. Although graph-based methods [58]–[60] have been proposed to share information between views from the perspective of preserving local manifold structure, they force the representations from different views to share the same structural distribution and correlation, which not only reduces the flexibility of information sharing but also is susceptible to unfriendly views and conveys unreliable information.

We propose a novel hypergraph-based information sharing model to solve the above-mentioned problems by dividing the information carried by LVs into structure information and view-dependent information. The structure information, such as the label distribution, can be shared to convey reliable information. The view-dependent information, such as the difference between representations in the spectral dimension, can be used to preserve the specificity of LVs. In detail, considering that all LVs have the same label distribution and share the common label matrix $\mathbf{Y}$, we use label information to construct hyperedges so that representations from the same class are located in the same hyperedge. Hence, different LVs share the same set of hyperedges $\mathcal{E} = \{\mathcal{E}_1, \ldots, \mathcal{E}_C\}$ and incident matrix $\mathbf{H} \in \mathbb{R}^{n \times C}$. The degree e_c of the hyperedge $\mathcal{E}_c (1 \leq c \leq C)$ is equal to the number of samples belonging to the class c. Note that the bands of representations from different LVs are different. That is to say, although the representations from the same class belong to the same hyperedge, the compactness between them varies for different LVs and is view dependent, which is utilized to preserve the specificity of LVs. This can be done by automatically assigning different weights to the same hyperedge related to various LVs. Hence, from the perspective of preserving manifold structure, a hypergraph autolearning-based information sharing model can be modeled as

$$\begin{aligned} &\min \sum_{v=1}^{V} \sum_{c=1}^{C} \sum_{i,j=1}^{n} h_{ic} h_{jc} \left\| \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_i^{(v)} - \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_j^{(v)} \right\|_2^2 \frac{\left(w_c^{(v)}\right)^r}{e_c} \\ &\text{s.t. } \mathbf{w}_{(v)} \geq 0 \quad \text{and} \quad \sum_{c=1}^{C} w_c^{(v)} = 1 (v = 1, \ldots, V) \end{aligned} \tag{13}$$

w.r.t $\mathbf{w}_{(v)} = [w_1^{(v)}, \ldots, w_C^{(v)}]^{\mathrm{T}} (v = 1, \ldots, V)$, where V represents the number of LVs and r is a parameter to avoid trivial solution. Let $\mathbf{q}_{(v)} = (q_c^{(v)}) \in \mathbb{R}^C$ denote the weight vector of hyperedges of the vth LV, where the weight $q_c^{(v)}$ of the hyperedge $\mathcal{E}_c$ is equal to $(w_c^{(v)})^r$. Then, the degree vector $\mathbf{v}_{(v)} = (v_c^{(v)}) \in \mathbb{R}^n$ of vertexes can be further calculated by (2).

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We can draw two conclusions by analyzing the objective (13). First, all LVs share the same incident matrix $\mathbf{H}$, which indicates that hypergraphs associated with different LVs have the same hyperedge set, thus further conveying reliable structure information (namely, label distribution) between LVs. Second, the hyperedge weights are view dependent. The optimization of $\mathbf{w}_{(v)}$ in Section III-B.2 shows that the weight of any hyperedge of the vth LV is determined by the compactness of the representations belonging to it. The specificity of different LVs is preserved in this way. Therefore, the model (13) provides a reliable and flexible way to share information between LVs. Actually, the traditional graph-based information sharing, which makes samples from different LVs share the same similarity measure, is a special form of the model (13). In detail, if all LVs share the same hyperedge weight vector $\mathbf{w}$, the model (13) is equivalent to the following graph-based manner [58]–[60]:

$$\min \sum_{v=1}^{V} \sum_{i,j=1}^{n} \left\| \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_i^{(v)} - \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_j^{(v)} \right\|_2^2 \mathbf{S}_{ij} \tag{14}$$

where $\mathbf{S}_{ij} = \sum_{c=1}^{C} h_{ic} h_{jc} (\mathbf{w}_c / e_c)$ denote the similarity between the ith and jth representations.

4) LvaHAl Model and Its Online Version: In the above-mentioned process, each local projection matrice only measures the importance of bands related to the corresponding LV. The remaining key is how to integrate the band information mined by these local projection matrices to select the desired bands from a global perspective. Based on the assumption that the importance of bands is continuous, we learn a consensus matrix $\mathbf{W} \in \mathbb{R}^{d \times C}$ to accomplish this task. This assumption states that if a band is important at a certain LV or time, it is equally important at other LVs or times. Hence, the consensus matrix $\mathbf{W}$ is regularized as follows:

$$\min_{\mathbf{W}} \sum_{v=1}^{V} \alpha_{(v)} \| \mathbf{W}_{(v)} - \mathcal{P}_v(\mathbf{W}) \|_F^2 \tag{15}$$

where the nonnegative parameters $\{\alpha_{(v)}\}_{v=1}^{V}$ reflect the importance of different LVs. The matrix $\mathbf{W}$ is considered to capture the importance of all bands at a global level. The sign $\mathcal{P}_v(\mathbf{W})$ represents the selection operator acting on $\mathbf{W}$ and returns a new matrix with the same number of rows and columns as matrix $\mathbf{W}_{(v)}$. The rows of the returned matrix are identical to the rows in matrix $\mathbf{W}$ that measure the same subset of bands as the rows in matrix $\mathbf{W}_{(v)}$. Finally, we incorporate the formulas (12), (13), and (15) together and propose the following LvaHAl model:

$$\begin{aligned} &\min \mathcal{H}\left(\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}, \mathbf{w}_{(v)}\}_{v=1}^{V}, \mathbf{W}\right) \\ &\quad = \sum_{v=1}^{V} \mathcal{F}(\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}) + \sum_{v=1}^{V} \alpha_{(v)} \| \mathbf{W}_{(v)} - \mathcal{P}_v(\mathbf{W}) \|_F^2 \\ &\quad + \beta \sum_{v=1}^{V} \sum_{c=1}^{C} \sum_{i,j=1}^{n} \left\| \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_i^{(v)} - \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_j^{(v)} \right\|_2^2 \frac{h_{ic} h_{jc} \left(w_c^{(v)}\right)^r}{e_c} \\ &\text{s.t. } \mathbf{M}_{(v)} \geq 0, \mathbf{w}_{(v)} \geq 0, \text{ and } \sum_{c=1}^{C} w_c^{(v)} = 1, \quad v = 1, \ldots, V \end{aligned} \tag{16}$$

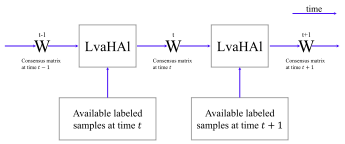


Fig. 4. Online version of the LvaHAl model.

where β is used to control the importance of the hypergraph-based information sharing item.

When new labeled samples are available, the traditional BS methods need to retrain the model in conjunction with current and historical samples to select the appropriate bands. Therefore, they cannot deal with dynamic samples efficiently. Thanks to the assumption that the importance of bands is continuous, the LvaHAl mode (16) can be easily and seamlessly extended by repeatedly revising the consensus matrix to handle this situation. Specifically, let $\overset{t}{\mathbf{W}}$ denote the consensus matrix learned by processing training samples at time t. The matrix $\overset{t}{\mathbf{W}}$ can be used as an initialization of $\overset{t+1}{\mathbf{W}}$ when new labeled samples are available at time $t+1$. Based on the current samples, the LvaHAl model can then derive the consensus matrix $\overset{t+1}{\mathbf{W}}$ by alternately revising $\overset{t}{\mathbf{W}}$ and all local projection matrices. Hence, the historical information of bands is encoded into the updated consensus matrix $\overset{t+1}{\mathbf{W}}$. This process can be repeated as long as new samples are available, as shown in Fig. 4. Hence, the LvaHAl model has the ability to process dynamic samples and can be viewed as an online BS model.

B. Optimization

As seen from the model (16), there are three groups of variables that need to be optimized: $\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}\}_{v=1}^{V}$ related to processing isolated pixels and performing BS, $\{\mathbf{w}_{(v)}\}_{v=1}^{V}$ related to reflecting the specificity of LVs, and $\mathbf{W}$ related to measuring bands from a global perspective. An alternate iterative algorithm is proposed to optimize these variables. In other words, we fix two groups of variables and optimize the rest one alternatively until convergence.

1) Optimizing $\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}\}_{v=1}^{V}$ *by Fixing* $\mathbf{W}$ *and* $\{\mathbf{w}_{(v)}\}_{v=1}^{V}$*:* With fixed $\{\mathbf{w}_{(v)}\}_{v=1}^{V}$ and $\mathbf{W}$, due to the independence between LVs, the variables $\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}\}_{v=1}^{V}$ can be solved in parallel. Taking the vth LV as an example, the optimization objective associated with $\mathbf{W}_{(v)}$, $\mathbf{b}_{(v)}$, and $\mathbf{M}_{(v)}$ can be rewritten as

$$\begin{aligned} &\min \mathcal{H}(\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}) \\ &\quad = \mathcal{F}(\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}) + \alpha_{(v)} \| \mathbf{W}_{(v)} - \mathcal{P}_v(\mathbf{W}) \|_F^2 \\ &\quad + \beta \sum_{c=1}^{C} \sum_{i,j=1}^{n} \left\| \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_i^{(\mathbf{v})} - \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_j^{(\mathbf{v})} \right\|_2^2 \frac{h_{ic} h_{jc} \left(w_c^{(v)}\right)^r}{e_c} \\ &\text{s.t. } \mathbf{M}^{(v)} \geq 0. \end{aligned} \tag{17}$$

Considering

$$\begin{aligned} &\sum_{c=1}^{C} \sum_{i,j=1}^{n} \left\| \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_i^{(\mathbf{v})} - \mathbf{W}_{(v)}^{\mathrm{T}} \mathbf{x}_j^{(\mathbf{v})} \right\|_2^2 \frac{h_{ic} h_{jc} \left(w_c^{(v)}\right)^r}{e_c} \\ &\qquad = 2\mathrm{tr}\left(W_{(v)}^{\mathrm{T}} \mathbf{X}_{(v)} \mathbf{L}_{(v)} \mathbf{X}_{(v)}^{\mathrm{T}} \mathbf{W}_{(v)}\right) \end{aligned} \tag{18}$$

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Algorithm 1 Algorithm to Solve Problem (17)

Input: Sample matrix $\mathbf{X}_{(v)}$, label matrix $\mathbf{Y}$, Initial $\mathbf{W}_{(v)}$, consensus matrix $\mathbf{W}$, incident matrix $\mathbf{H}$, the weight set of hyperedges $\mathbf{w}_{(v)}$, and parameters r, λ, β, and $\alpha_{(v)}$.
Initialize $g_i^{(v)} = 1(i = 1, \ldots, n)$ and $\mathbf{M}_{(v)} = \mathbf{0}$.
Calculate $\mathbf{L}_{(v)}$ by Eq. (4).
Repeat
Update the diagonal matrix $\mathbf{P}_{(v)}$, where the i-th diagonal element is equal to $\frac{1}{2\|\mathbf{w}_i^{(v)}\|}$.
Update $\mathbf{W}_{(v)}$ by Eq. (22).
Update $\mathbf{b}_{(v)}$ by Eq. (21).
Update $\mathbf{M}_{(v)}$, where the i-th column $\mathbf{m}_i^{(v)}$ is calculated by Eq. (24).
Update $g_i^{(v)}(i = 1, \ldots, n)$ by Eq. (20).
Until convergence
Output: $\mathbf{W}_{(v)}$, $\mathbf{b}_{(v)}$, and $\mathbf{M}_{(v)}$.

where $\mathbf{L}_{(v)}$ is the Laplacian matrix of the hypergraph derived from the vth LV, (17) can be solved by optimizing the following objective:

$$\begin{aligned}\min_{\mathbf{W}_{(v)},\mathbf{b}_{(v)},\mathbf{M}_{(v)}\geq 0} &\sum_{i=1}^{n} g_i^{(v)} \left\|\mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{x}_i^{(v)} + \mathbf{b}_{(v)} - \mathbf{z}_i^{(v)}\right\|_2^2 \\ &+ \lambda \mathrm{tr}\left(\mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{P}_{(v)}\mathbf{W}_{(v)}\right) + \alpha_{(v)}\|\mathbf{W}_{(v)} - \mathcal{P}(\mathbf{W})\|_F^2 \\ &+ 2\beta \mathrm{tr}\left(W_{(v)}^{\mathrm{T}}\mathbf{X}_{(v)}\mathbf{L}_{(v)}\mathbf{X}_{(v)}^{\mathrm{T}}\mathbf{W}_{(v)}\right) \end{aligned} \tag{19}$$

with $\mathbf{z}_i^{(v)} = \mathbf{y}_i + \mathbf{y}_i \odot \mathbf{m}_i^{(v)}$ and

$$g_i^{(v)} = \begin{cases} 1, & \text{if } \left\|\mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{x}_i^{(v)} + \mathbf{b}_{(v)} - \mathbf{z}_i^{(v)}\right\|_2^2 \leq \epsilon_{(v)} \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

The ith diagonal element of the diagonal matrix $\mathbf{P}_{(v)}$ is equal to $\frac{1}{2\left\|\mathbf{w}_i^{(v)}\right\|}$. When $\mathbf{M}_{(v)}$ is fixed, by setting the derivative of (19) w.r.t. $\mathbf{b}_{(v)}$ to zero, we have

$$\mathbf{b}_{(v)} = \frac{1}{\mathbf{1}^{\mathrm{T}}\mathbf{g}_{(v)}}\left(\mathbf{Z}_{(v)}\mathbf{g}_{(v)} - \mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{X}_{(v)}\mathbf{g}_{(v)}\right) \tag{21}$$

with $\mathbf{g}_{(v)} = [g_1^{(v)}, \ldots, g_n^{(v)}]^{\mathrm{T}}$ and $\mathbf{Z}_{(v)} = [\mathbf{z}_1^{(v)}, \ldots, \mathbf{z}_n^{(v)}]$. Substituting (21) into (19) and then setting the derivative of (19) w.r.t. $\mathbf{W}_{(v)}$ to zero, we have

$$\mathbf{W}_{(v)} = \mathbf{S}_{(v)}^{-1}\left(\mathbf{X}_{(v)}\mathbf{U}_{(v)}\mathbf{Z}_{(v)}^{\mathrm{T}} + \alpha_{(v)}\mathcal{P}_v(\mathbf{W})\right) \tag{22}$$

with $\mathbf{S}_{(v)} = \mathbf{X}_{(v)}(\mathbf{U}_{(v)} + 2\beta\mathbf{L}_{(v)})\mathbf{X}_{(v)}^{\mathrm{T}} + \lambda\mathbf{P}_{(v)} + \alpha_{(v)}\mathbf{I}$ and $\mathbf{U}_{(v)} = \mathrm{diag}(\mathbf{g}^{(v)}) - (1/\mathbf{1}^{\mathrm{T}}\mathbf{g}_{(v)})\mathbf{g}_{(v)}\mathbf{g}_{(v)}^{\mathrm{T}}$. When $\mathbf{W}_{(v)}$ and $\mathbf{b}_{(v)}$ are fixed, $\mathbf{M}_{(v)}$ can be solved by separately optimizing the following objective for each $\mathbf{m}_i^{(v)}$:

$$\min_{\mathbf{m}_i^{(v)}\geq 0} \left\|\mathbf{y}_i \odot \left(\mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{x}_i^{(v)} + \mathbf{b}_{(v)}\right) - \mathbf{1} - \mathbf{m}_i^{(v)}\right\|_2^2. \tag{23}$$

We can easily obtain the closed solution of $\mathbf{m}_i^{(v)}$ as follows:

$$\mathbf{m}_i^{(v)} = \left(\mathbf{y}_i \odot \left(\mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{x}_i^{(v)} + \mathbf{b}_{(v)} - \mathbf{y}_i\right)\right)_+. \tag{24}$$

Detailed procedure is summarized in Algorithm 1.

2) Optimizing $\{\mathbf{w}_{(v)}\}_{v=1}^V$ by Fixing $\mathbf{W}$ and $\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}\}_{v=1}^V$: When other variables are fixed, the model (16) is separable for $\{\mathbf{w}_{(v)}\}_{v=1}^V$. Similarly, for the vth LV, the model (16) w.r.t. $\mathbf{w}_{(v)}$ changes to

$$\begin{aligned} &\min_{\mathbf{w}_{(v)}} \sum_{c=1}^{C} l_c^{(v)}(w_c^{(v)})^r \\ &\text{s.t. } w_c^{(v)} \geq 0 \quad \text{and} \quad \sum_{c=1}^{C} w_c^{(v)} = 1 \end{aligned} \tag{25}$$

with

$$l_c^{(v)} = \sum_{i,j=1}^{n} \left\|\mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{x}_i^{(v)} - \mathbf{W}_{(v)}^{\mathrm{T}}\mathbf{x}_j^{(v)}\right\|_2^2 \frac{h_{ic}h_{jc}}{e_c}. \tag{26}$$

The closed solution of $w_c^{(v)}$, which can be obtained by solving the Lagrangian function of (25), is provided as follows:

$$w_c^{(v)} = \frac{\left(l_c^{(v)}\right)^{\frac{1}{1-r}}}{\sum_{c=1}^{C}\left(l_c^{(v)}\right)^{\frac{1}{1-r}}}. \tag{27}$$

3) Optimizing $\mathbf{W}$ by Fixing $\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}\}_{v=1}^V$ and $\{\mathbf{w}_{(v)}\}_{v=1}^V$: The Model (16) consisting of the terms associated with $\mathbf{W}$ has the following form:

$$\min_{\mathbf{W}} \sum_{v=1}^{V} \alpha_{(v)}\|\mathbf{W}_{(v)} - \mathcal{P}(\mathbf{W})\|_F^2. \tag{28}$$

Equation (28) can be expressed equivalently as

$$\min_{\mathbf{W}} \sum_{v=1}^{V} \alpha_{(v)}\|\hat{\mathbf{W}}_{(v)} - \mathbf{C}_{(v)}\mathbf{W}\|_F^2 \tag{29}$$

with $\hat{\mathbf{W}}_{(v)} \in \mathbb{R}^{d\times C}$ and $\mathbf{C}_{(v)} \in \mathbb{R}^{d\times d}$. For matrix $\hat{\mathbf{W}}_{(v)}$, the rows that measure the same bands as the rows of matrix $\mathbf{W}_{(v)}$ are identical to the rows of matrix $\mathbf{W}_{(v)}$, and the values of the other rows are zero. The diagonal elements of the diagonal matrix $\mathbf{C}_{(v)}$ corresponding to the nonzero rows in $\hat{\mathbf{W}}_{(v)}$ are 1, and the other diagonal elements are 0. By setting the derivative of (29) w.r.t. $\mathbf{W}$ to zero, we can obtain the closed solution of $\mathbf{W}$ as follows:

$$\mathbf{W} = \left(\sum_{v=1}^{V} \alpha_{(v)}\mathbf{C}_{(v)}^{\mathrm{T}}\mathbf{C}_{(v)}\right)^{-1}\left(\sum_{v=1}^{V} \alpha_{(v)}\mathbf{C}_{(v)}^{\mathrm{T}}\hat{\mathbf{W}}_{(v)}\right). \tag{30}$$

The complete procedure of solving LvaHAl is summarized in Algorithm 2. When the consensus matrix $\mathbf{W}$ is obtained, we evaluate the score of the ith band by means of $\left\|\mathbf{w}^i\right\|_2$. Finally, the bands with top-p scores are selected.

As seen from the procedure of solving LvaHAl in Algorithm 2, we solve the optimization of three sets of variables $\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}\}_{v=1}^V$, $\{\mathbf{w}_{(v)}\}_{(v=1)}^V$, and $\mathbf{W}$ alternatively. The computational complexity solving each set of variables is listed as follows.

1) The first set of variables $\{\mathbf{W}_{(v)}, \mathbf{b}_{(v)}, \mathbf{M}_{(v)}\}_{v=1}^V$ is solved iteratively by Algorithm 1. For vth LV, when $\mathbf{b}_{(v)}$ and $\mathbf{M}_{(\mathbf{v})}$ are fixed, the computational complexity of deriving $\mathbf{W}_{(v)}$ is $\mathcal{O}(n^2\ d_v + nd_v^2 + d_v^3 + nCd_v + Cd_v^2)$. Similarly, the computational complexity in solving $\mathbf{b}_{(v)}$ and $\mathbf{M}_{(v)}$

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Algorithm 2 LvaHAl for BS

1: **Input**: samples matrices $\{\mathbf{X}_{(v)}\}_{v=1}^{V}$, label matrix $\mathbf{Y}$, incident matrix $\mathbf{H}$, parameters r, λ, β, and $\{\alpha_{(v)}\}_{v=1}^{V}$.
2: Initialize the weights of all hyperedges as $\frac{1}{C}$ for all local views. Initialize $\mathbf{W} = \mathbf{0}$.
3: **Repeat**
4: Update $\mathbf{W}_{(v)}$, $\mathbf{b}_{(v)}$, and $\mathbf{M}_{(v)}$ $(v = 1, \ldots, V)$ by Alg. 1.
5: Update $\mathbf{w}_{(v)}(v = 1, \ldots, V)$, where i-th element $w_i^{(v)}$ is calculated by Eq. (27).
6: Update $\mathbf{W}$ by Eq. (29).
7: **Until convergence**
8: **Output**: $\mathbf{W}$.

is nCd_v. Considering that n is greater than C, in general, the total computational complexity is $\mathcal{O}(k_1 \sum_{v=1}^{V}(n^2 + nd_v + d_v^2)d_v)$, where k_1 is the number of iterations of Algorithm 1.

2) The second set of $\{\mathbf{w}_{(v)}\}_{v=1}^{V}$ aims to update the hyperedge weights. It is updated element by element for each LV. The total computational complexity is $\mathcal{O}(n^2\ C\sum_{v=1}^{V} d_v)$.
3) The third variable $\mathbf{W}$ has closed form solution (30). Since $\mathbf{C}_{(v)}$ is a diagonal matrix, the complexity of solving the inverse of $\sum_{v=1}^{V}(\alpha_{(v)}\mathbf{C}_{(v)}^{\mathrm{T}}\mathbf{C}_{(v)})$ is at $\mathcal{O}(d)$. Hence, the computational complexity of updating $\mathbf{W}$ is $\mathcal{O}(VCd^2)$.

In summary, the total computational complexity of Algorithm 2 is $\mathcal{O}(k_2(\sum_{v=1}^{V}(k_1(n^2 + nd_v + d_v^2) + n^2\ C)d_v + VCd^2))$, where k_2 is the number of iterations of Algorithm 2.

IV. Experiments

A. Data Sets

In the experimental part, three hyperspectral data sets are used to verify the performance of the LvaHAl model. A detailed description of these data sets is shown in the following.

The first data set is a well-known University of Pavia image acquired with the Reflective Optics System Imaging Spectrometer (ROSIS) sensor having 1.3-m spatial resolutions. This data set consists of 610 × 340 pixels with 9 classes, in which each pixel has 115 spectrum bands ranging from 0.43 to 0.86 μm. After removing 12 noisy bands, the remaining 103 bands are used for BS and classification.

The second data set, the Pavia Center image, was also obtained by the ROSIS sensor over Pavia, northern Italy. Hence, it has the same spatial and spectral resolutions as the first data set. In this data set, 1096 × 715 pixels from nine classes are included. After noisy spectra are removed, the number of available bands is 102 for the experiments.

The third data set is the Salinas image that covers an area located in Salinas Valley, CA, USA. This image was obtained by the Airborne Visible/Infrared Imaging Spectrometer, having 3.7-m spatial resolutions. It consists of 517 × 217 pixels with 16 classes. When the 20 noise bands (108–112, 154–167, and 224) in terms of water absorption are removed, 204 bands are retained for experimental analysis.

Table II lists the number of training and test pixels for three data sets, respectively.

B. Experimental Setup

This section consists of three components, which are evaluation criteria, comparison algorithms, and two classifiers for classification, respectively.

1) Evaluation Criteria: In order to demonstrate the effectiveness of LvaHAl as fairly as possible, the following evaluation criteria are considered: the classification performance, McNemar's test, and the computational time. In our experiments, the classification performance is evaluated by two indicators. One is the overall accuracy (OA) that is defined as the percentage of correctly classified pixels to all test pixels. The other is the kappa coefficient that represents the percentage of correctly classified pixels by the number of agreements that could be expected due to chance alone. Besides, McNemar's test T is a widely used indicator for comparing two algorithms statistically, which is computed as follows [61]:

$$T = \frac{f_{12} - f_{21}}{\sqrt{f_{12} + f_{21}}}. \tag{31}$$

The sign $f_{ab}(a \neq b \in \{1, 2\})$ denotes the number of samples classified correctly by algorithm a and incorrectly by algorithm b. Algorithm a is statistically superior to algorithm b if $T > 1.96$ and vice versa if $T < -1.96$.

2) Comparison Algorithms: We compare our method with the following eight state-of-the-art BS methods, namely, the linearly constrained minimum variance (LCMV) with the band correlation/dependence minimization (LCMV-BCM/BDM) [22], LCMV with the band correlation/dependence constraint (LCMV-BCC/BDC) [22], fast and robust principal component analysis on Laplacian graph (FRPCALG) [48], FS [33], robust feature selection (RFS) [39], SVM with recursive feature elimination (SVM-RFE) [62], SSVM [38], and BS based on the Hilbert–Schmidt independence criterion (HSIC) [34]. Besides, the results obtained using all original bands are also reported as a baseline.

Among these methods, LCMV-BCC/BDC and LCMV-BCM/BDM belong to the classical BS methods that examine the relationship between bands. FS is a classical discriminative BS method. FRPCALG first obtains the low-rank approximation of the band matrix to mitigate the effects of noise and further utilizes graph-based Laplacian constraint to evaluate bands. For HL-based SVM-RFE and SSVM, they are designed to deal with the binary-class case. We use a one-versus-rest strategy to add the band scores of each binary-class subproblems and get the final bands with high scores for the multiclass case. Besides, the gaussian kernel is adopted in SVM-RFE. For RFS, the $l_{2,1}$-norm-based regression loss function is adopted to enforce robustness to outliers, and the $l_{2,1}$-norm constraint is added to select bands simultaneously. HSIC formulates BS as an optimization of the least square regression regularized by l_1-norm to select discriminative bands. For FRPCALG, SVM-RFE, SSVM, RFS, and our method, the regularization parameters are selected from

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TABLE II
NUMBER OF TRAINING AND TESTING PIXELS OF DIFFERENT DATA SETS

No	The first dataset			The second dataset			The third dataset		
	Class	Training	Testing	Class	Training	Testing	Class	Training	Testing
C1	Asphalt	30	6601	Water	30	65941	Brocoli_green_weeds_1	30	1979
C2	Meadows	30	18619	Trees	30	7568	Brocoli_green_weeds_2	30	3696
C3	Gravel	30	2069	Asphalt	30	3060	Fallow	30	1946
C4	Trees	30	3034	Self-Blocking Bricks	30	2655	Fallow_rough_plow	30	1364
C5	Painted metal sheets	30	1315	Bitumen	30	6554	Fallow_smooth	30	2648
C6	Bare Soil	30	4999	Tiles	30	9218	Stubble	30	3929
C7	Bitumen	30	1300	Shadows	30	7257	Celery	30	3549
C8	Self-Blocking Bricks	30	3652	Meadows	30	42796	Grapes_untrained	30	11241
C9	Shadows	30	917	Bare Soil	30	2833	Soil_vinyard_develop	30	6173
C10							Corn_senesced_green_weeds	30	3248
C11							Lettuce_romaine_4wk	30	1038
C12							Lettuce_romaine_5wk	30	1897
C13							Lettuce_romaine_6wk	30	886
C14							Lettuce_romaine_7wk	30	1040
C15							Vinyard_untrained	30	7238
C16							Vinyard_vertical_trellis	30	1777
	Total	270	42506		270	147912		480	53649

the candidate set $\{10^{-7}, 10^{-5}, 10^{-3}, 10^{-1}, 10^{1}, 10^{3}\}$. For HSIC, the regularization parameter is selected from the set $\{10^{-11}, 10^{-10}, \ldots, 10^{4}\}$ to select a suitable number of spectral bands. Besides, the number of neighbors is selected from the set $\{5, 10, 15, 20\}$ for FRPCALG, and the value of r is selected from the set $\{1.5, 2.0, 2.5, 3.0\}$ for our method. To facilitate comparison, we set $\alpha_{(\upsilon)}$s to be the same and replace them with α for all LVs in our method. Similar to [56], we determine the value $\epsilon_{(\upsilon)}$ of the υth local view by selecting 10% samples with the largest noise in the first five iterations.

3) Classifier: For classification, all samples are first represented by the selected bands for different BS methods. Then, classifiers are trained on the constructed training samples. Finally, the constructed test samples are classified by the learned classifiers. Two classic classifiers, including k-nearest neighbors (KNN) [63] and SVM [64], [65], are adopted in our experiments. The optimal k value of KNN is obtained from the set $\{2, 3, 5, 6, 10, 15\}$. For SVM, a linear kernel is used, and the involved penalty factor is selected from the set $\{10^{-3}, 10^{-2}, \ldots, 10^{5}\}$. The above-mentioned adjustable parameters are determined by five-fold cross-validation.

C. Experimental Results

For given training samples, we obtain several LVs by generating LVs (see Section III-A.1). The values of $\tau_{\min}$ and $\tau_{\max}$ are set to $1/3$ and $2/3$ in our experiments, respectively. Since training samples are randomly selected, we, therefore, report an average of 10 running results for different BS methods. All experiments are performed using MATLAB R2017b on a laptop with 8.0-GB RAM memory.

1) First Data Set: For the first data set, Fig. 5 plots the curves of OA and Kappa obtained by KNN and SVM. We can observe three facts from Fig. 5. First, the indicators OA and Kappa have similar trends with the increase of available bands for different BS methods. Second, on the KNN classifier, all methods are able to achieve better or comparable classification results than the baseline with a small number of bands, which

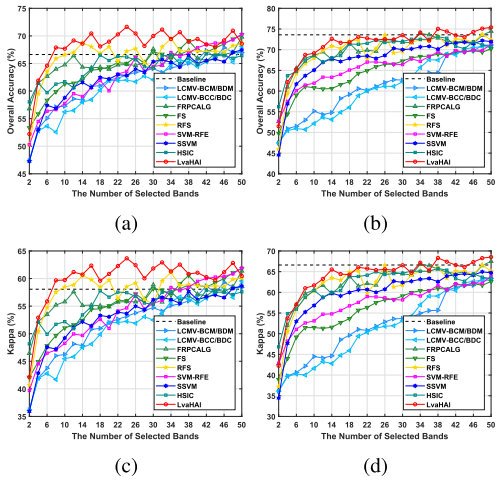


Fig. 5. Classification results of the first data set obtained by KNN and SVM with the bands selected by different BS methods. (a) OA obtained by KNN. (b) OA obtained by SVM. (c) Kappa obtained by KNN. (d) Kappa obtained by SVM.

is not true on the SVM classifier. For example, only LvaHAl, RFS, and FRPCALG obtain kappa coefficient values on SVM that are comparable to the baseline [see Fig. 5(d)]. Third, there are large-performance fluctuations between different BS methods, even if a large number of bands are selected. Taking KNN as an example, the difference in OA between LvaHAl and LCMV-BCC/BDC is as high as 5% when nearly half of the bands (e.g., 48 bands) are available.

In terms of classification performance, the advantages of different BS methods can be evaluated in two ways: 1) the minimum number of bands required to be better the baseline and 2) the highest classification accuracy obtained with the selected bands. It is not difficult to observe from Fig. 5 that our method performs significantly better than other comparison algorithms in the above-mentioned two aspects. As shown in Fig. 5(a), on the KNN classifier, LvaHAl achieves better OA 67.89% than the baseline with only 8 bands and the highest OA 71.65% with 24 bands. Among its competitors, the OA value 68.02% superior to the baseline is obtained by RFS

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TABLE III

AVERAGE MCNEMAR'S TEST BETWEEN OUR METHOD AND COMPARISON ALGORITHMS ON THE KNN CLASSIFIER FOR THE FIRST DATA SET

Bands	8	16	24	32	36
LCMV-BCM/BDM	39.79	37.05	35.59	30.72	24.98
LCMV-BCC/BDC	55.73	45.55	38.25	31.71	20.27
FRPCALG	16.15	25.47	27.98	26.80	17.61
FS	30.05	26.95	27.48	18.17	12.47
RFS	8.88	9.24	19.98	13.23	14.95
SVM-RFS	43.89	36.75	33.30	25.02	18.49
SSVM	38.64	36.49	34.84	23.91	26.33
HSIC	25.63	25.00	23.15	19.56	24.12

using 10 bands, and the optimal OA 70.28% is obtained by SVM-RFE using 50 bands. Similar phenomena also occur on the SVM classifier. Hence, we can infer from these facts that our method can obtain better classification results with fewer bands, which indicates that it is sensitive to discriminative bands. In other words, the LvaHAl method makes it easier to identify bands that are useful for distinguishing different surface objects. Finally, we provide the average McNemar's test between LvaHAl and its competitors in Table III when the number of available bands is set to 8, 16, 24, 32, and 36, respectively. All values of McNemar's test in Table III are greater than 1.96, which indicates that LvaHAl is superior to these comparison algorithms in a statistical sense.

2) Second Data Set: Fig. 6 shows the performance trends of different BS methods on the second data set when the number of selected bands changes. As shown in Fig. 6, as the available bands increase, the values of OA and Kappa increase gradually and then tend to be stable on both classifiers. For example, for these BS methods, the maximum change in OA is only 0.67% on KNN when the number of selected bands is not less than 36. By observing Fig. 6(b) and (d), it can be seen that on SVM, LvaHAl not only achieves better classification performance than the baseline with fewer bands but also obtains the highest OA and Kappa. For example, the OA 94.76% superior to baseline and best OA 95.32% are obtained using 16 and 40 bands provided by LvaHAl, respectively [see Fig. 6(b)]. Fig. 6(a) and (c) shows that the optimal and suboptimal classification results (e.g., 93.82% and 93.74% for OA) on the KNN classifier are obtained by FS and LvaHAl using 46 and 26 bands, respectively. Although the FS method yields a 0.08% accuracy advantage, it selects 20 more bands than our method, which effectively increases the time for subsequent processing (e.g., training of classifiers). In addition, we can observe from Fig. 6 that the performance of LvaHAl is significantly better than that of SVM-RFE and SSVM, which may be attributed to the robustness of our method to isolated pixels. Finally, we report the average McNemar's test between LvaHAl and comparison methods in Table IV on the KNN classifier for a different number of bands. The fact that all values in Table IV are greater than 1.96 indicates that our method is statistically better than its competitors.

3) Third Data Set: In Fig. 7, the performance curves versus selected bands on two classifiers are plotted for different BS methods. We can observe that LvaHAl outperforms the existing BS methods in two aspects: the minimum number of bands required to be better the baseline and the highest classification accuracy obtained with the selected bands. In particular, only our method achieves better classification results than the baseline on SVM, which implies that LvaHAl can identify discriminative bands without loss of information on the Salinas image. Hence, we can reasonably claim that our proposed LvaHAl method is an effective discriminative BS model for insufficient labeled samples. Table V lists the average McNemar's test between LvaALa and comparison algorithms on KNN, which supports our statement earlier.

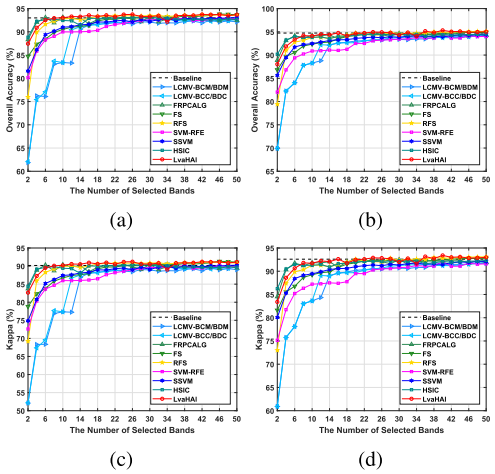


Fig. 6. Classification results of the second data set obtained by KNN and SVM with the bands selected by different BS methods. (a) OA obtained by KNN. (b) OA obtained by SVM. (c) Kappa obtained by KNN. (d) Kappa obtained by SVM.

TABLE IV

AVERAGE MCNEMAR'S TEST BETWEEN OUR METHOD AND COMPARISON ALGORITHMS ON THE KNN CLASSIFIER FOR THE SECOND DATA SET

Bands	8	12	16	20	24
LCMV-BCM/BDM	103.63	105.95	35.96	33.16	33.79
LCMV-BCC/BDC	99.50	46.16	35.89	30.10	30.75
FRPCALG	15.57	9.32	29.41	10.66	10.29
FS	44.11	35.22	34.25	16.59	13.23
RFS	10.36	7.04	11.48	4.46	5.64
SVM-RFS	49.19	43.71	47.85	35.04	30.26
SSVM	36.75	33.46	31.03	25.79	22.09
HSIC	3.86	15.92	11.09	10.73	14.52

TABLE V

AVERAGE MCNEMAR'S TEST BETWEEN OUR METHOD AND COMPARISON ALGORITHMS ON THE KNN CLASSIFIER FOR THE THIRD DATA SET

Bands	18	28	44	46	50
LCMV-BCM/BDM	28.50	17.57	16.84	10.38	13.56
LCMV-BCC/BDC	31.39	18.66	18.43	13.00	16.52
FRPCALG	1.56	3.11	5.16	1.30	8.89
FS	29.26	20.63	15.51	13.85	16.01
RFS	5.17	2.00	6.22	3.42	7.14
SVM-RFS	23.91	9.55	9.55	8.82	11.96
SSVM	24.56	8.96	9.46	6.57	11.82
HSIC	4.18	1.94	6.04	4.65	9.19

Finally, we visualize the convergence behavior of Algorithm 2 on the three data sets in Fig. 8 and show the average time consumed by various BS methods in Table VI. In this process, there are two convergence conditions for the proposed algorithms: 1) the relative change of objective value in two successive iterations is less than a predefined threshold (e.g., 0.001) and 2) the number of iterations exceeds the maximum (e.g., 100). Fig. 8 shows that Algorithm 2 can

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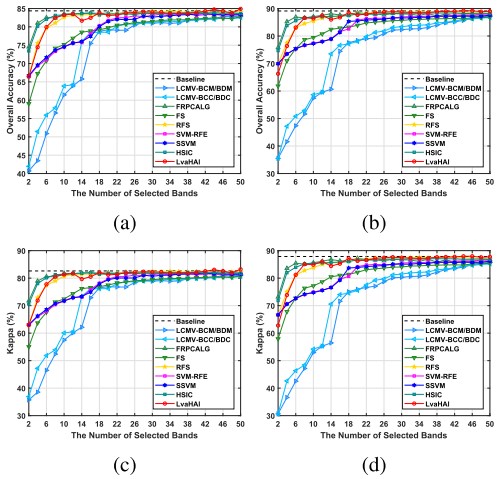


Fig. 7. Classification results of the third data set obtained by KNN and SVM with the bands selected by different BS methods. (a) OA obtained by KNN. (b) OA obtained by SVM. (c) Kappa obtained by KNN. (d) Kappa obtained by SVM.

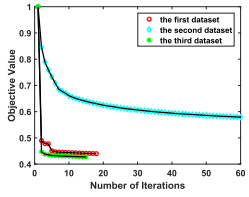


Fig. 8. Convergence behavior of LvaHAl on different data sets.

achieve local convergence after several iterations for different data sets. In addition, Table VI shows that LvaHAl does not have significant advantages in terms of computation time when compared with other competitors, which can be analyzed from the following three aspects. First, LvaHAl uses a more complex loss function to process isolated pixels. Second, LvaHAl indirectly increases the number of training samples by means of LVs. Third, different combinations of parameters affect the convergence of the LvaHAl model.

D. Analysis of the Parameters

The LvaHAl model includes four parameters λ, β, α, and r. The values of the first three parameters are limited to the set $\{10^{-7}, 10^{-5}, \ldots, 10^{3}\}$, and the value of r is in the set $\{1.5, 2.0, 2.5, 3.0\}$. We verify the effect of each parameter on all data sets when other parameters are fixed, as shown in Fig. 9. We can see from Fig. 9(a)–(c) that the sensitivity of LvaHAl to parameters λ, β, and α is different on three data sets. For the first and second data sets, our method is robust to the parameter β and sensitive to the parameters λ and α. Fortunately, LvaHAl achieves good and stable classification accuracy for smaller values of parameters λ and α (e.g., no more than 0.1). That is to say, LvaHAl is robust to λ and α in a local range. A similar situation can also be observed in the third data set. Fig. 9(d) shows that the classification performance is less affected by the parameter r, which indicates that LvaHAl is robust to this parameter. Hence, taking relatively smaller values of these parameters is a reasonable choice in practical applications.

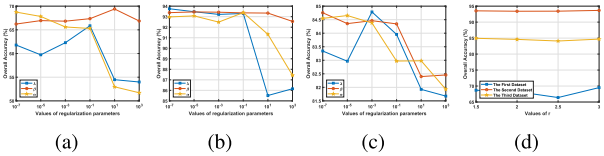


Fig. 9. Classification performance with regard to different parameters of the LvaHAl model on the KNN classifier. (a)–(c) represent the impact of parameters λ, β, and α on three data sets, respectively. (a) First data set. (b) Second data set. (c) Third data set. (d) Impact of parameter r on three data sets.

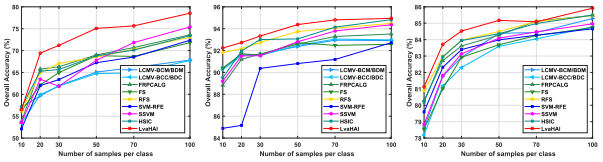


Fig. 10. Taking KNN as an example, the classification accuracy OA varies with the training samples on the three data sets for different BS methods. (a) First data set. (b) Second data set. (c) Third data set.

E. Analysis of Training Samples

We show the effect of training samples on classification accuracy when the number of selected bands is set to 24, 16, and 50 on three data sets, respectively. The number of training samples corresponding to each class is limited to the set $\{10, 20, 30, 50, 70, 100\}$. Fig. 10 displays the OA curves of these data sets on KNN when the number of training samples is different. First of all, it can be observed that with the increase in available training samples, the performance of most BS methods increases markedly and steadily. Second, in most cases, the LvaHAl method is able to achieve better classification results than comparison methods. For example, on the second data set, when the number of training samples per class is 10, LvaHAl still achieves a performance advantage of 0.42% to 3.40% over other competitors. This indicates that the way we designed to handle the lack of labeled samples is reasonable and effective. Based on these phenomena, we can reasonably conclude that even in the case of rare labeled samples, our method can also obtain a subset of bands containing as much discriminative information as possible.

F. Analysis of the Impact of Local Views

LvaHAl solves the conflict between insufficient labeled samples and high-dimensional bands in the process of evaluating bands by means of LVs. Apparently, two aspects of LVs are affected by the values of $\tau_{\min}$ and $\tau_{\max}$ in our method, namely, the total number of LVs and the band dimension of each LV. Therefore, we explore the impact of LVs by varying $\tau_{\min}$ and $\tau_{\max}$. In the experiments, combinations of $\tau_{\min}$ and $\tau_{\max}$ are limited to the set $\{(1/5, 1/4), (1/5, 1/3), \ldots, (1/2, 3/4), (2/3, 3/4)\}$. The element $(1/5, 1/4)$ indicates that the values of $\tau_{\min}$ and $\tau_{\max}$ are $1/5$ and $1/4$ when generating LVs, respectively. Taking the first data set as an example, Fig. 11 shows the effect of LVs on the performance of LvaHAl on KNN. By comparing Fig. 11(a)–(c), it can be seen that the effect of LVs on classification accuracy gradually weakens with the increase in selected bands. Besides, there is no obvious correlation

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TABLE VI

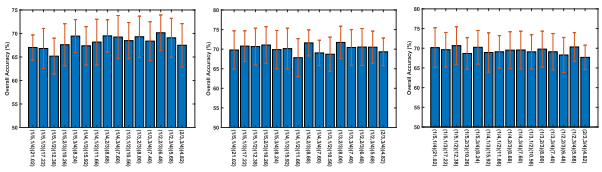
COMPUTATIONAL TIMES (IN SECONDS) OF DIFFERENT BS METHODS ON ALL DATA SETS

Bands	The first dataset					The second dataset					The third dataset				
	8	16	24	32	36	8	12	16	20	24	18	28	44	46	50
LCMV-BCM/BDM	39.44	39.45	39.45	39.45	39.45	192.09	192.09	192.09	192.09	192.09	93.59	93.59	93.59	93.59	93.59
LCMV-BCC/BDC	41.37	41.37	41.37	41.37	41.37	215.43	215.43	215.43	215.43	215.43	94.79	94.79	94.79	94.79	94.79
FRPCALG	3.55	3.73	3.54	3.52	3.54	1.90	3.59	1.88	3.52	3.62	8.92	10.91	0.32	0.31	10.58
FS	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.05	0.05	0.04	0.04	0.04
RFS	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.10	0.10	0.10	0.10	0.10
SVM-RFE	1.40	1.43	1.33	0.45	0.90	0.46	0.47	0.48	0.43	0.46	3.23	14.43	14.67	15.16	15.21
SSVM	24.89	24.42	24.75	25.52	25.07	23.02	23.10	23.74	23.74	24.66	0.25	0.25	0.31	0.32	0.35
HSIC	1.93	2.26	1.96	2.24	1.99	2.18	2.15	2.20	2.23	2.21	11.06	11.05	11.03	11.04	11.06
LvaHAl	1.29	1.63	1.69	3.65	1.56	7.56	5.70	6.26	6.89	7.70	9.34	7.58	23.13	21.28	18.27

TABLE VII

PERFORMANCE OF THE ONLINE LVAHAL AND LVAHAL METHODS IN TERMS OF ACCURACY OBTAINED BY KNN AND TIME WITH THE INCREASING NUMBER OF TRAINING SAMPLES. THE NUMBER OF SELECTED BANDS FOR THREE DATA SETS IS SET TO 10, 10, AND 44, RESPECTIVELY

Number of training samples for each class	The first dataset				The second dataset				The third dataset			
	Online LvaHAl		LvaHAl		Online LvaHAl		LvaHAl		Online LvaHAl		LvaHAl	
	OA	time	OA	time	OA	time	OA	time	OA	time	OA	time
30 (0)	70.69	1.08	71.72	1.36	93.35	5.38	92.80	5.52	82.39	23.37	82.22	25.13
60 (30)	72.89	0.71	74.96	5.55	94.03	5.16	93.50	17.02	84.31	16.30	83.84	71.23
90 (30)	75.79	0.72	76.44	11.56	94.34	4.86	94.20	27.02	85.94	23.32	85.89	147.99
150 (60)	77.17	3.89	78.91	27.76	94.40	13.64	94.78	53.63	86.05	76.50	85.39	399.89
210 (60)	79.50	3.75	79.72	50.01	94.83	14.25	95.40	96.47	86.38	57.58	86.06	743.91
300 (90)	80.82	8.99	80.44	103.63	95.52	26.91	95.47	151.24	86.60	160.40	85.91	1225.94



(a) (b) (c)

Fig. 11. Average OA and corresponding standard deviation related to the first data set for different number of LVs on KNN. $(1/5, 1/4)(21.02)$ represents that the average number of LVs is 21.02 when the values of $\tau_{\min}$ and $\tau_{\max}$ are set to $1/5$ and $1/4$, respectively. (a) Ten selected bands. (b) Twenty selected bands. (c) Thirty selected bands.

between performance variance and LVs. Fortunately, those LVs that lead to better average OA generally make the performance fluctuation smaller. These facts indicate that obtaining good LVs is helpful to improve classification performance and stability, which will be the focus of our next step. Moreover, Fig. 11 indicates that too few or too many bands associated with LVs are more likely to cause poor accuracy. However, the reasons for these two situations may be different. Too few bands may cause loss of information corresponding to LV, further leading to a biased estimation of the local projection matrix. Conversely, too many bands exacerbate the conflict between the lack of labeled samples and high-dimensional bands when evaluating bands, which implies that considering the high dimensionality of bands is necessary, and our approach is effective in solving the above-mentioned conflict.

G. Analysis of the Effectiveness of Online LvaHAl

In this section, we verify the effectiveness of online LvaHAl based on the framework, as shown in Fig. 4. We compare the performance of the online LvaHAl and LvaHAl methods in terms of accuracy and time when the number of training samples for each class increases by 30, 60, or 90, as shown in Table VII. In this Table VII, 60 (30) represents that the current number of training samples per class is 60, which is obtained by adding 30 samples to the corresponding class of the previous samples. Table VII shows that with the gradual increase in available samples, the online LvaHAl and LvaHAl methods can achieve similar OA, which indicates that the former does not lose valuable information. In addition, when new samples are available, the online version has higher learning efficiency because there is no need to train the model from scratch. For example, when the number of training samples per class is increased to 300, the time consumed by LvaHAl is about 11.53 times that of the online LvaHAl while they achieve comparable OA (e.g., 80.82% and 80.44%) for the first data set. As a result, the above-mentioned facts prove that our proposed online version of LvaHAl is valid.

V. CONCLUSION

In this article, we propose a novel BS method called LvaHAl. Our approach first randomly divides the band space to obtain LVs. For each LV, a capped HL with the row-sparsity constraint is adopted to deal with the isolated pixels and pick the bands that are optimal in corresponding LV. Then, we propose a novel hypergraph learning model, which generalizes the traditional graph-based model, to provide reliable and flexible information sharing between LVs. This model allows all LVs to share the same hypergraph structure while automatically adjusting view-dependent hyperedge weights in the optimization process. Finally, based on the assumption that

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the importance of bands is continuous, we learn a consensus matrix to obtain the optimal band subset from a global perspective. Moreover, an online version of our method is developed. Experimental results show that our method can obtain discriminative band subsets and good classification performance. In addition, with the increasing number of available samples, its online version can effectively save calculation time without degrading classification performance.

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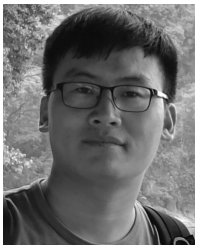
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